

PROCESSING AND ANALYSIS OF DATA

Data processing is done for making comparisons and analysis.

Processing implies editing, coding, classification and tabulation of collected data so that they are amenable to analysis.

Analysis refers to computation of certain measures and searching for patterns of relationship that exist among data-groups.

PROCESSING OPERATIONS

1. Editing is done to detect errors and omissions and to correct these when possible.

Purpose of editing is to assure accuracy, consistency with other facts, uniformity, and arranged to facilitate coding and tabulation.

2. Coding refers to process of assigning numerals or other symbols to data to facilitate classification /categorization.

3. Classification: Large volume of raw data is put into homogeneous groups to draw meaningful relationships.

Classification is done in two ways.

(i) *Classification according to attributes:*

(ii) *Classification according to class-intervals:*

Size of class interval:

$$i = R / (1 + 3.3 \log N)$$

where

i = size of class interval;

R = Range (i.e., difference between Max. – Min.)

N = Number of items to be grouped.

Open-ended intervals in the overall frequency distribution.

Types of class interval

<i>Exclusive type class intervals:</i>	<i>Inclusive type class intervals</i>
10–20	11–20
20–30	21–30
30–40	31–40
40–50	41–50
For continuous data	For discrete data

(iii) Frequency of each class?

This can be done either by tally sheets or by mechanical aids, or data processing software using computer.

4. Tabulation:

Tabulation is the process of summarising raw data and displaying the same in compact form in rows and columns for analysis.

Tabulation is essential because

1. It conserves space
2. facilitates the process of comparison
3. facilitates summation of items and detection of errors and omissions

4. provides a basis for various statistical computations

Tabulation may also be classified as simple and complex tabulation.

Simple tabulation gives information about one or more groups of independent questions, whereas complex tabulation shows division of data in two or more categories.

Complex tabulation

two-way tables -> two inter-related characteristics of data

three-way tables -> three interrelated characteristics of data

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k-way table (manifold table) -> levels of k categorical variables

Simple table S(i) -> i represents class

	B.Tech	M.Tech	MBA	Total
Students	295	95	55	445

Two way table S(i, j) -> i represents class and j represents branch

	B.Tech	M.Tech	PhD	Total
Civil	72	30	18	120
Electrical	74	34	12	120
Electronics	77	25	15	117
Mechanical	72	36	10	118
Total	295	125	55	475

Three way table S(i, j, k) -> i, j, k represents class, branch and gender

	B.Tech		M.Tech		PhD		Total	
	M	F	M	F	M	F	M	F
	1	2	3	4	5	6		
1. Civil	44	28	12	18	9	9	65	55
2. Electrical	41	33	17	17	7	5	65	55
3. Electronics	42	35	12	13	7	8	61	56
4. Mechanical	43	29	20	16	6	4	69	49
Total	170	125	61	64	29	26	260	215

1. Every table should have a clear, concise and adequate title
2. Every table should be given a distinct number to facilitate easy reference.
3. Column headings (captions) and the row headings (stubs) of table should be clear.
4. Units of measurement under each heading should be indicated.
5. Explanatory footnotes, if any, should be placed directly beneath the table.
6. Source or sources of data must be indicated just below the table.
7. The columns may be numbered to facilitate reference.
8. Those columns whose data are to be compared should be kept side by side.

9. It is generally considered better to approximate figures before tabulation.
10. Miscellaneous and exceptional items, if any, should be placed in the last row of table.

ELEMENTS/TYPES OF ANALYSIS

Analysis involves

- Estimating the parameters of the population (Descriptive analysis)
- Testing of hypotheses for drawing inferences (Inferential analysis also known as statistical analysis)

DESCRIPTIVE ANALYSIS is may be in respect of one variable, or two variables, or more than two variables (described as multivariate analysis).

Univariate Analysis

Population parameters

- (1) measures of central tendency -> Mean, Median, Mode
- (2) measures of dispersion -> Variance, Standard Deviation
- (3) measures of asymmetry (skewness)

Bivariate Analysis

Correlation analysis -> How two variables are related.

Causal analysis -> how one or more variables affect changes in another variable. This analysis can be termed as regression analysis.

Multivariate analysis -> simultaneously analyse more than two variables

- (a) *Multiple regression analysis*: one dependent variable which is a function of two or more independent variables.
- (b) *Multiple discriminant analysis*: used for categorical DV on basis of some attribute.
- (c) *Multivariate analysis of variance (or multi-ANOVA)*:

INFERENTIAL ANALYSIS is concerned with testing hypotheses for desired level of significance.

Inferential statistics are also known as **sampling statistics**

Testing of Hypothesis

1. z test 2. t test	3. Chi square test 4. Sign Test 5. Run Test 6. Kruskal-Wallis test
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